

Qwen3.5-9B UK: religion — every mitigation against the baseline on the same rows

Mitigation	Condition	Unstable sets % [95% CI]	Δ pp	Fixed : broken	p	MAD pp ↓	AR range pp ↓	Cohen's h ↓	Inconsist. % ↓	Hire %	Utility % ↑	Mean FS	Attr. mentioned %
baseline (no mitigation)	implicit	21.33 [16.66, 26.23]				4.43	18.89	0.13	5.11	17.23	77.28	0.57	5.70
SFT (LoRA, merged)	implicit	3.56 [1.78, 5.56] ♦	-17.78 ♦	87 : 7	1e-18	0.42 (4.43) ♦	1.33 (18.89) ♦	0.02 (0.13) ♦	0.86 (5.11) ♦	11.53 (17.23)	75.58 (77.28)	0.65 (0.57)	0.00 (5.70) ♦
LEACE (layer 16)	implicit	26.16 [20.00, 33.19] (own baseline 33.76, n=237)	-7.59	42 : 24	0.04	6.59 (4.43)	31.87 (18.89)	0.14 (0.13)	5.18 (5.11)	27.36 (17.23)	79.78 (77.28)	0.60 (0.57)	3.90 (5.70)
prompt: structured_rubric	implicit	13.78 [10.00, 18.22]	-7.56	48 : 14	2e-05	3.29 (4.43)	11.78 (18.89)	0.12 (0.13)	3.51 (5.11)	10.69 (17.23)	74.15 (77.28)	0.54 (0.57)	2.89 (5.70)
prompt: recruiter_guidelines	implicit	17.56 [13.55, 22.00]	-3.78	28 : 11	9e-03	3.71 (4.43)	15.56 (18.89)	0.12 (0.13)	4.27 (5.11)	12.99 (17.23)	74.47 (77.28)	0.58 (0.57)	4.96 (5.70)
prompt: ignore_personal_info	implicit	19.56 [15.33, 23.78]	-1.78	30 : 22	0.33	3.29 (4.43)	15.33 (18.89)	0.09 (0.13)	4.49 (5.11)	23.41 (17.23)	79.95 (77.28) ♦	0.61 (0.57)	0.91 (5.70)
prompt: counterfactual_invariance	implicit	19.78 [15.78, 24.22]	-1.56	34 : 27	0.44	3.59 (4.43)	13.11 (18.89)	0.09 (0.13)	4.96 (5.11)	22.42 (17.23)	78.32 (77.28)	0.59 (0.57)	1.16 (5.70)
prompt: reasoning	implicit	20.71 [16.22, 25.50]	-0.67	19 : 16	0.74	4.52 (4.43)	19.11 (18.89)	0.14 (0.13)	4.99 (5.11)	14.47 (17.23)	75.60 (77.28)	0.53 (0.57)	4.42 (5.70)
scrub: llm	implicit	21.83 [17.11, 26.73]	+0.45	38 : 40	0.91	1.95 (4.43)	9.62 (18.89)	0.04 (0.13)	5.63 (5.11)	22.64 (17.23)	78.98 (77.28)	0.60 (0.57)	1.04 (5.70)
prompt: zero_shot_cot	implicit	22.67 [18.44, 27.33]	+1.33	15 : 21	0.41	4.55 (4.43)	19.11 (18.89)	0.13 (0.13)	5.31 (5.11)	14.49 (17.23)	76.02 (77.28)	0.58 (0.57)	5.46 (5.70)
prompt: fairness_constitution	implicit	22.89 [18.67, 27.11]	+1.56	68 : 75	0.62	2.85 (4.43)	15.56 (18.89)	0.06 (0.13)	5.21 (5.11)	38.20 (17.23)	79.53 (77.28)	0.63 (0.57)	0.12 (5.70)
prompt: second_pass_verification	implicit	33.56 [28.22, 39.34]	+12.22	24 : 79	5e-08	2.72 (4.43)	15.56 (18.89)	0.09 (0.13)	7.93 (5.11)	23.11 (17.23)	75.41 (77.28)	0.58 (0.57)	1.78 (5.70)
scrub: lexical*	implicit	0.67 [0.00, 1.56]	-20.67	96 : 3	5e-25	0.04 (4.43)	0.22 (18.89)	0.00 (0.13)	0.07 (5.11)	29.14 (17.23)	81.63 (77.28)	0.63 (0.57)	0.00 (5.70)
baseline (no mitigation)	explicit	10.89 [8.22, 14.00]				1.59	6.22	0.04	2.77	31.16	81.09	0.62	0.10
scrub: llm	explicit	0.67 [0.00, 1.56] ♦	-10.22 ♦	49 : 3	1e-11	0.16 (1.59) ♦	0.67 (6.22) ♦	0.01 (0.04) ♦	0.15 (2.77) ♦	31.01 (31.16)	80.54 (81.09)	0.62 (0.62)	0.00 (0.10) ♦
SFT (LoRA, merged)	explicit	1.33 [0.44, 2.44]	-9.56	48 : 5	7e-10	0.22 (1.59)	0.89 (6.22)	0.01 (0.04)	0.37 (2.77)	12.20 (31.16)	75.65 (81.09)	0.65 (0.62)	0.00 (0.10) ♦
LEACE (layer 16)	explicit	11.11 [7.80, 14.80] (own baseline 17.41, n=270)	-6.30	41 : 24	0.05	3.03 (1.59)	13.23 (6.22)	0.07 (0.04)	3.28 (2.77)	44.15 (31.16)	73.95 (81.09)	0.62 (0.62)	0.17 (0.10)
prompt: recruiter_guidelines	explicit	8.22 [5.78, 10.89]	-2.67	32 : 20	0.13	0.99 (1.59)	3.78 (6.22)	0.03 (0.04)	1.80 (2.77)	25.04 (31.16)	80.74 (81.09)	0.64 (0.62)	0.05 (0.10)
prompt: structured_rubric	explicit	8.22 [5.78, 10.89]	-2.67	37 : 25	0.16	1.07 (1.59)	4.89 (6.22)	0.03 (0.04)	2.07 (2.77)	20.81 (31.16)	78.05 (81.09)	0.57 (0.62)	0.02 (0.10)
prompt: reasoning	explicit	9.33 [6.89, 12.00]	-1.56	25 : 18	0.36	1.27 (1.59)	4.22 (6.22)	0.04 (0.04)	2.15 (2.77)	28.27 (31.16)	81.56 (81.09)	0.60 (0.62)	0.05 (0.10)
prompt: zero_shot_cot	explicit	10.44 [7.56, 13.56]	-0.44	24 : 22	0.88	1.51 (1.59)	6.67 (6.22)	0.04 (0.04)	2.35 (2.77)	28.96 (31.16)	82.10 (81.09) ♦	0.64 (0.62)	0.07 (0.10)
prompt: ignore_personal_info	explicit	12.00 [8.67, 15.56]	+1.11	19 : 24	0.54	2.05 (1.59)	8.89 (6.22)	0.05 (0.04)	2.77 (2.77)	34.02 (31.16)	79.56 (81.09)	0.62 (0.62)	0.00 (0.10) ♦
prompt: counterfactual_invariance	explicit	12.22 [9.33, 15.33]	+1.33	24 : 30	0.50	1.89 (1.59)	7.78 (6.22)	0.06 (0.04)	2.91 (2.77)	32.10 (31.16)	79.41 (81.09)	0.62 (0.62)	0.00 (0.10) ♦
prompt: fairness_constitution	explicit	14.48 [11.14, 18.00]	+3.56	29 : 45	0.08	2.54 (1.59)	11.33 (6.22)	0.07 (0.04)	3.04 (2.77)	40.52 (31.16)	79.38 (81.09)	0.66 (0.62)	0.00 (0.10) ♦
prompt: second_pass_verification	explicit	20.00 [16.22, 24.00]	+9.11	15 : 56	1e-06	2.19 (1.59)	10.44 (6.22)	0.05 (0.04)	4.84 (2.77)	31.53 (31.16)	78.44 (81.09)	0.63 (0.62)	0.02 (0.10)
scrub: lexical*	explicit	1.11 [0.22, 2.00]	-9.78	47 : 3	4e-11	0.13 (1.59)	0.44 (6.22)	0.00 (0.04)	0.37 (2.77)	28.81 (31.16)	81.46 (81.09)	0.63 (0.62)	0.00 (0.10)

Values: mitigated run, baseline in brackets, both on exactly the rows the run evaluated. Unstable sets: share with its own 95% CI (bootstrap over candidates); Δ pp: change against the baseline; bold coloured Δ = p < 0.05 (exact sign test on fixed vs broken sets, uncorrected). A run with parse failures is compared on its complete sets only; its own baseline on those sets is shown under its value. Italic ♦ = best value in the column within its condition (real mitigations only; hire rate and FS have no better direction).

Fixed : broken — each counterfactual set (one candidate-job pair, one condition, all attribute variants) paired with the same set at baseline. Fixed: unstable at baseline (the attribute alone changed the decision), stable after the mitigation. Broken: stable at baseline, unstable after — the mitigation introduced a new dependence on the attribute. Δ unstable sets = (broken – fixed) / sets; e.g. 58 : 57 means the method fixes about as many sets as it breaks.

* Lexical scrubbing is an oracle upper bound, not a comparable mitigation: it removes the attribute with the study's own injection templates, so every variant becomes character-for-character the attribute-free CV (100% of rows) and the model sees one identical prompt — zero instability by construction. Real CVs do not use our templates; read this row only as the ceiling. LLM scrubbing, which must find the attribute itself, is the realistic version.